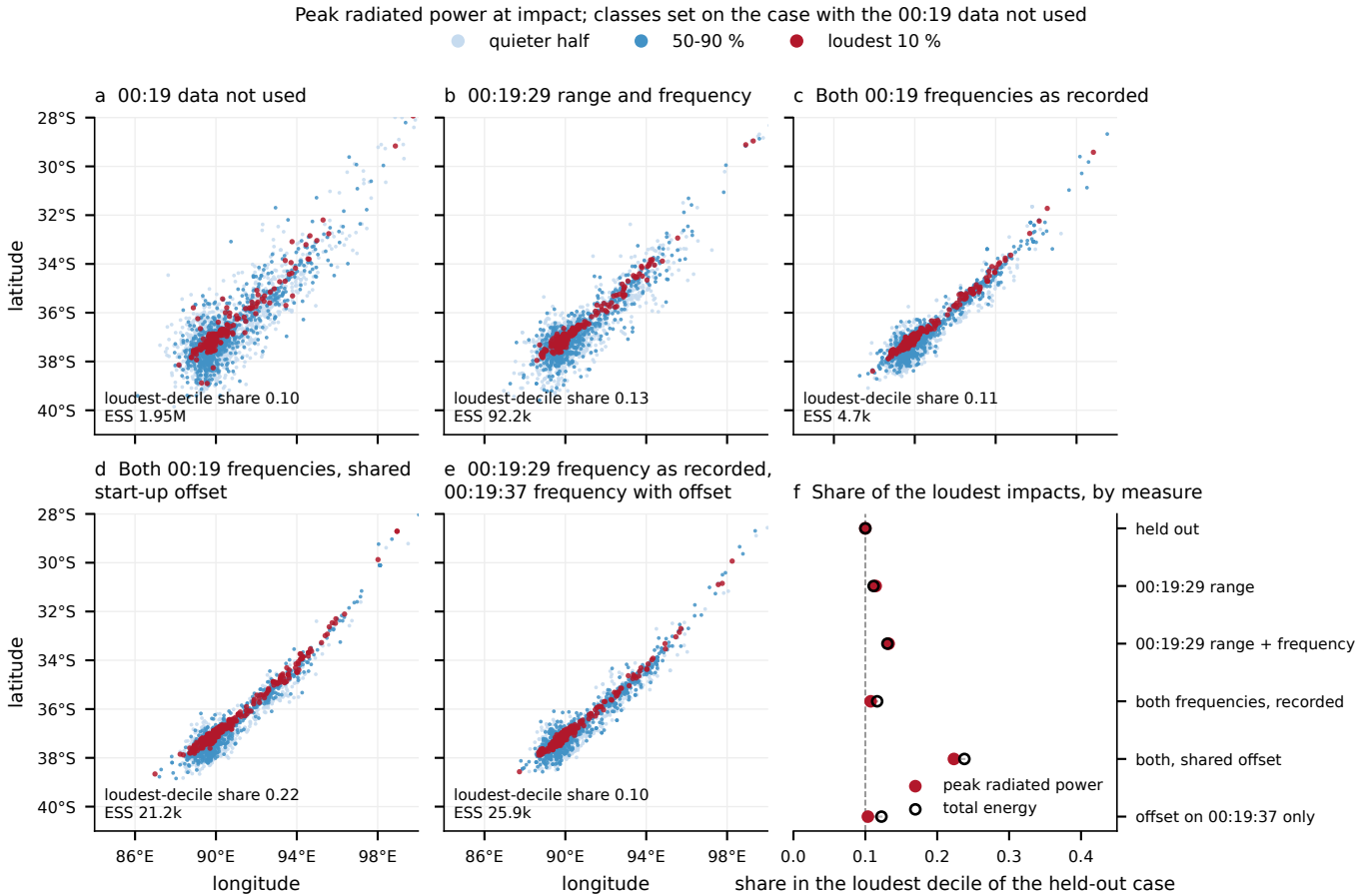



Note 1: Each dot is one likely place where the aircraft hit the ocean. The colour shows how loud the impact was, as peak radiated power. The classes come from the case that does not use the 00:19 data. Panel f shows how much of each case is in the loudest tenth, by peak power and by total energy. If the 00:19 data do not change loudness, the share is 0.10 (dashed line). The flight model for a descent without control is not calibrated, so the loudness and energy are not physical predictions.

Note 2: NOT A PHYSICAL PREDICTION: The uncontrolled-arm dynamics are not calibrated: the Boeing leave-one-out gate failed in all three rounds. Impact energy, attitude, flight-path angle and energy-transfer duration from this arm are a labelled sensitivity on uncalibrated dynamics, not a physical prediction. End of flight on core run J (accepted as a pass by project ruling, 16 Oct 2026): 4 route families with 24/18/16/16 seeds x 100,000 hand-off particles at 00:11; 16 children x 4 descents per particle (waypoints), 8 x 4 (others); family weights 0.9982/0.0013/0.0003/0.0002; binary mh370-eof-rj1 (sha256 19031f41). Round-3 uncontrolled-descent laws; spiral bank limit uniform 52-90 deg; dive to surface p = 0.5. Log-on causes C1 fuel exhaustion (Erlang lag) + C2 other (1/W, W = 00:10:58-01:15:56 UTC, silent), q = 1/3; survival factor (no in-flight breakup before 00:19:37) applied; transmission at 00:19 imposed. No treatment uses the 00:19:37 range. Treatments: D0' held out, D1 R600 BTO, D2 R600 BTO+BFO, D3 both BFOs raw, D4 both BFOs + shared start-up offset (Holland), D5 R600 BFO raw + R1200 BFO with offset U[17,130] Hz. Families: A1 = code 1, A2 = code 2, B = codes 3+6 (B uncontrolled about 0.001). PROVISIONAL: piloted-turn model, INTERIM pitch-down. Classes: weighted quantiles of D0' peak radiated power, median 4.2e+08 W, top decile 7.2e+10 W (total energy top decile 11.0 GJ), tau90_v1 sidecar. Each map: 3,000 impacts drawn by weight from a mass-preserving systematic resample of 10,000 rows per seed per treatment; all four strata. ESS: Kish effective impacts pooled over seeds.